

Problem Set 2

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Operating Systems

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Task 2

```
huehuehussain@DESKTOP-I8H9IR2: ~/xv6-labs-2025
-fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
huehuehussain@DESKTOP-I8H9IR2:~/xv6-labs-2025$ make
make: 'kernel/kernel' is up to date.
huehuehussain@DESKTOP-I8H9IR2:~/xv6-labs-2025$ ./grade-lab-syscall
make: 'kernel/kernel' is up to date.
== Test answers-syscall.txt ==
answers-syscall.txt: FAIL
Cannot read answers-syscall.txt
== Test sandbox_mask == sandbox_mask: OK (4.6s)
== Test sandbox_fork == sandbox_fork: OK (0.6s)
== Test sandbox_path == sandbox_path: FAIL (1.0s)
Number of appearances of 'xv6'
got:
4
expected:
12
QEMU output saved to xv6.out.sandbox_path
== Test sandbox_most == sandbox_most: OK (0.9s)
== Test sandbox_minus == sandbox_minus: FAIL (1.0s)
Number of appearances of 'hello!'
got:
2
expected:
3
QEMU output saved to xv6.out.sandbox_bigpath
== Test attack == attack: FAIL (0.8s)
...
init: starting sh
$ secret beDwiHA0
$ attack
$ attack
$ qemu-system-riscv64: terminating on signal 15 from pid 21195 (make)
MISSING '^beDwiHA0.*'
QEMU output saved to xv6.out.attack
== Test time ==
time: FAIL
Cannot read time.txt
Score: 15/45
huehuehussain@DESKTOP-I8H9IR2:~/xv6-labs-2025$
```

(a) Make Grade(Task 2)

Task 3

```
huehuehussain@DESKTOP-I8H9IR2: ~/xv6-labs-2025
ltn-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free ^
-fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-st
ack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/ka
lloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel
/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kerne
l/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_dis
k.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.s
ym
make[1]: Leaving directory '/home/huehuehussain/xv6-labs-2025'
== Test answers-syscall.txt ==
answers-syscall.txt: FAIL
    Cannot read answers-syscall.txt
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (4.9s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (1.1s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (1.5s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (0.6s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.0s)
    (Old xv6.out.sandbox_bigpath failure log removed)
== Test attack ==
$ make qemu-gdb
attack: FAIL (1.1s)
...
    init: starting sh
    $ secret QgaMBkH3
    $ attack
    $ attack
    $ qemu-system-riscv64: terminating on signal 15 from pid 24367 (make)
MISSING '^QgaMBkH3.*'
    QEMU output saved to xv6.out.attack
== Test time ==
time: FAIL
    Cannot read time.txt
Score: 25/45
make: *** [Makefile:371: grade] Error 1
huehuehussain@DESKTOP-I8H9IR2:~/xv6-labs-2025$
```

(a) Make Grade(Task 3)

Task 4

```
huehuehussain@DESKTOP-I8H9IR2: ~/xv6-labs-2025
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/virtio_disk.o kernel/virtio_disk.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/start.o kernel/start.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/console.o kernel/console.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/printf.o kernel/printf.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/uart.o kernel/uart.c
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plc.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printf.o kernel/uart.o kernel/spinlock.o
riscv64-linux-gnu-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
huehuehussain@DESKTOP-I8H9IR2:~/xv6-labs-2025$ ./grade-lab-syscall attack
make: 'kernel/kernel' is up to date.
== Test attack == attack: OK (7.9s)
huehuehussain@DESKTOP-I8H9IR2:~/xv6-labs-2025$
```

(a) Make Grade(Task 4)